Extended Abstract

Offline RL for Adaptive Vocabulary Selection in Conversational Language Tutoring

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Motivation A conversational language tutor must, on every turn, choose which mix of new, in-progress, and review vocabulary to weave into its response. Existing language-learning products handle this with hand-coded spaced-repetition rules (SM-2, FSRS) that schedule isolated flashcards and offer no mechanism for embedding review in dialogue. We treat the problem as sequential decision making and ask whether reinforcement learning can recover scheduling structure that fixed heuristics cannot express, under the constraint that policy learning must be *offline*: no student-in-the-loop training is allowed in any deployable product.

Method We decompose each turn into two stages. An RL policy selects a structured *directive* $(n_{\text{new}}, n_{\text{active}}, n_{\text{review}})$ summing to 5 words (21 discrete actions), and a frozen LLM (Claude Sonnet) realizes that directive as a Japanese dialogue turn. This decomposition keeps the LLM out of the training loop entirely. The reward combines coverage growth, a naturalness penalty calibrated against an LLM judge, and a session-boundary retention bonus. We train IQL and CQL on a rebalanced multi-behavior-policy offline buffer with PPO-distilled trajectories, with PPO as an online upper bound and behavior cloning of PPO as an ablation isolating how much of offline-RL's value comes from learning under deployment-realistic data versus simply imitating a good policy.

Implementation The simulator implements an FSRS-style retrievability $R(t) = (1 + t/(9c_f S))^{-1}$ over 500 JLPT N5/N4/N3 words. State is 2122-dimensional (4 features per word, plus two session-position globals). Offline RL uses 100k-step training, $\beta = 3$ AWR temperature, expectile $\tau = 0.7$, $100\times$ reward scaling to escape AWR weight collapse. PPO is a standard clipped-surrogate actor-critic. BC is a 256×256 MLP trained via cross-entropy on PPO state-action pairs. All training runs on Modal in parallel with WandB logging. Evaluation uses 20 seeds per policy, with post-gap retention (R> 0.5 after a 7-day gap) as the headline metric. Conversational naturalness and comprehension support are scored by a cross-family LLM judge (OpenAI gpt-4o-mini judging Claude Sonnet output) to mitigate same-family self-preference bias.

Results On post-gap retention with 95% bootstrap CIs over 20 eval seeds: IQL retains 85.8 words $[52.6, 119.0]$, CQL 81.3 $[49.9, 112.7]$, the strongest deployable baseline (Random) 18.0 $[15.2, 20.9]$, FSRS+New 4.4. Offline RL therefore achieves $4.8\times$ Random and $19\times$ FSRS+New. PPO (151.2) and BC of PPO (152.2) are statistically indistinguishable as a non-deployable reference. BC and PPO degrade in lockstep across six OOD simulator perturbations (correlation > 0.99), confirming BC's faithful imitation. The cross-family LLM judge scores all policies tightly in $[4.5, 5.0]$ on naturalness and $[4.25, 4.75]$ on comprehension. An IQL β sensitivity sweep (pooled over training seeds) peaks at $\beta = 5$ with 84.1 retained.

Discussion The BC≈PPO ablation reframes what offline RL contributes on this problem: once an online policy exists its behavior is recoverable by supervised cloning, so the value of IQL and CQL lies in being trainable on the deployable multi-behavior buffer of mediocre heuristic policies, not in extracting clever signal from expert trajectories. Reward design dominated algorithm choice in our experiments: the move from coverage-only to the three-term reward was the single largest jump in retention, larger than any algorithm or hyperparameter change.

Conclusion A hierarchical control architecture that separates RL (over structured linguistic constraints) from a frozen LLM (as the dialogue executor) is a viable design for tutor-style applications. Offline RL achieves order-of-magnitude retention gains over deployable heuristics while preserving conversational quality. The remaining gap to PPO measures how much harder it is to learn from a mixture of mediocre policies than from a single good one. Top-k per-word action spaces, comprehension-as-reward, and real-student calibration of the simulator are the natural next directions.

Offline RL for Adaptive Vocabulary Selection in Conversational Language Tutoring

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Abstract

We frame vocabulary selection in a conversational language tutor as a sequential decision problem: at every turn the tutor must choose a mix of new, in-progress, and review words that jointly maximizes long-horizon retention while preserving conversational naturalness. We instantiate this as a 500-word Japanese-vocabulary domain with an FSRS-based knowledge-tracing simulator, a 21-action structured directive space, and a frozen LLM realizing each directive as dialogue. We compare offline RL (IQL, CQL) on a rebalanced multi-behavior buffer to an online PPO oracle, a behavior-cloning of PPO ablation, and a battery of spaced-repetition baselines. Offline RL achieves $4.8\times$ the post-gap retention of the strongest deployable baseline (Random) and $19\times$ the FSRS+New spaced-repetition heuristic, at no cost in conversational naturalness. The BC-of-PPO ablation shows that, once an online policy is available, its behavior is recoverable by supervised cloning, meaning that the value of offline RL on this problem comes from learning under *deployment-realistic* multi-behavior data, not from sophisticated exploitation of expert trajectories. We additionally extend the LLM-judge to score conversational *comprehension support* as a separate axis, find that the high-retention learned policy (IQL) matches heuristic baselines on this axis, and report OOD generalization across simulator parameterizations.

1 Introduction

A conversational language tutor faces a control problem on every turn: it must select a small set of vocabulary items to weave into the response. The chosen items determine three quantities that the tutor jointly optimizes over a long horizon: (i) *coverage*, the cumulative fraction of the curriculum the learner has been exposed to; (ii) *retention*, the probability that previously-seen words remain recallable after time has passed; and (iii) *conversational naturalness*, whether the produced dialogue reads as a coherent exchange rather than a vocabulary drill. Existing language-learning products rely on hand-coded spaced-repetition schedulers (Anki/SuperMemo's SM-2 family (Wozniak, 1990), FSRS (Ye, 2024)). These work in flashcard settings but offer no mechanism for embedding review in dialogue and reason about isolated cards rather than jointly about turn composition.

This work asks whether reinforcement learning, particularly *offline* RL trained on logged behavioral data, can discover scheduling strategies that fixed heuristics cannot express. The setting demands offline rather than online RL because student-in-the-loop training is impractical: online algorithms would need to expose real learners to many sessions of an unconverged policy, which is unacceptable in any deployed tutoring product. We therefore train under a deployment-realistic constraint: only previously logged trajectories from existing heuristic policies are available, with no further environment interaction during policy learning.

Contributions.

- A hierarchical control architecture that decomposes turn composition into a 21-action structured directive (counts of new/active/review words) selected by RL, with a frozen LLM realizing the directive as natural dialogue. This keeps the LLM out of the training loop entirely.
- A reward function combining coverage growth, a naturalness penalty calibrated against LLM-judge ratings, and a session-end retention bonus that provides long-horizon credit assignment without an LLM in the training loop.
- Empirical comparison of IQL (Kostrikov et al., 2022), CQL (Kumar et al., 2020), PPO (Schulman et al., 2017), and behavior cloning of PPO on a 500-word Japanese-vocabulary simulator, with a cross-family LLM judge for naturalness and comprehension.
- A BC-of-PPO ablation isolating how much of offline-RL's value comes from learning under multi-behavior data versus simply imitating a good policy.
- An out-of-distribution simulator-parameter robustness study evaluating each method under forgetting-curve and difficulty perturbations.

2 Related Work

Spaced repetition. The Leitner box system (Leitner, 1972) and SuperMemo's SM-2 family (Wozniak, 1990) schedule isolated flashcards via fixed exponential-backoff rules. FSRS (Ye, 2024) learns the forgetting-curve parameters from review logs but retains the per-card abstraction. Settles and Meeder (2016) fit a half-life regression model on Duolingo review data; their domain is flashcards, not dialogue. None of these methods reason about composition: how to combine items into a single tutor turn, which is the central decision in conversational tutoring.

RL for tutoring. Rafferty et al. (2016) cast teaching as POMDP planning with a probabilistic student model, working on isolated problems with hand-specified dynamics. Most subsequent RL-for-tutoring work (Tarsouri and Samarati, 2023) continues this pattern: item-level decisions on problem-solving tasks, never on item selection mediated through a generative dialogue model.

Offline RL. IQL (Kostrikov et al., 2022) uses expectile regression to estimate the value of the in-support optimal action and advantage-weighted regression (Peng et al., 2019) to extract the policy; CQL (Kumar et al., 2020) adds a conservative penalty that pushes down Q-values on out-of-distribution actions. Both are designed for the static-buffer setting we operate in.

LLM control. RLHF (Ouyang et al., 2022) treats the LLM itself as the policy. We adopt the opposite decomposition: RL operates over a small structured action space (linguistic constraints), and the LLM is a frozen executor of those constraints. This is closer in spirit to hierarchical control than to end-to-end LLM fine-tuning.

Behavior cloning. BC (Pomerleau, 1989) is the natural ablation for any offline-RL claim: if a supervised model trained on expert trajectories matches the RL method, the RL machinery is not adding signal beyond imitation. We report BC-of-PPO results as a deliberate ablation.

3 Method

Simulator and environment. We instantiate a 500-word Japanese vocabulary curriculum spanning JLPT levels N5–N3 (35%/35%/30%). Each word has stability S_w, a difficulty $d_w \in [0, 10]$ scaled from JLPT level, and an FSRS-style retrievability $R_w(t) = (1 + t/(9 \cdot c_f \cdot S_w))^{-1}$, where t is days since the last exposure and c_f is a per-environment forgetting constant (default 1.0). On a Bernoulli-successful recall, stability grows multiplicatively in a difficulty-modulated factor; on failure it resets; first exposures seed $S_0 = 0.5$. Time advances $\sim$2 minutes per turn and one day between sessions.

State. $s \in \mathbb{R}^{2122}$ concatenates per-word (S, R, d, usage) for all 530 words plus two global session-position features (turn-in-session and session index).

Action. An action a is a directive $(n_{\text{new}}, n_{\text{act}}, n_{\text{rev}})$ specifying how many words of each category to target this turn. Counts sum to 5; this yields 21 discrete actions. The environment deterministically maps each directive to specific word IDs by retrievability-sorting active and review pools and difficulty-sorting the new pool. The frozen LLM then receives those word IDs and produces the actual turn text.

Reward. The reward r_t combines three terms:

$$r_t = \underbrace{\frac{\Delta R_{\text{total}}}{N}}_{\text{coverage growth}} - \underbrace{\alpha \cdot \max(0,\, n_{\text{new}} - 1)^2}_{\text{naturalness penalty}} + \underbrace{\lambda \cdot \mathbf{1}[t = T_{\text{sess}}] \cdot \frac{\#\{R_w > 0.5\}}{N}}_{\text{retention bonus}} \tag{1}$$

The coverage term provides dense per-turn signal; the naturalness penalty (calibrated $\alpha = 0.01$ against LLM-judge ratings) discourages multi-new-word turns; the retention bonus at session boundaries ($\lambda = 1.0$) provides long-horizon credit assignment without requiring an LLM in the training loop. We multiply the final reward by $100\times$ before training to escape advantage-weighted-regression weight collapse at the buffer's large state dimension.

IQL. We train an expectile-regression value function $V(s)$ at expectile $\tau = 0.7$ and a Q-function $Q(s, a)$ minimizing $(Q - r - \gamma V(s'))^2$. The policy is extracted via advantage-weighted regression with inverse-temperature β: $\pi(a \mid s) \propto \exp(\beta \cdot (Q(s, a) - V(s)))$. We compare $\beta \in \{1, 3, 5, 10, 30\}$.

CQL. Standard conservative Q-learning: we add a regularizer $\alpha_{\text{CQL}} \cdot (\log \sum_a \exp Q(s, a) - \mathbb{E}_{a\sim\mathcal{D}}[Q(s, a)])$ to the IQL critic loss, which pushes down Q-values for out-of-distribution actions. We use $\alpha_{\text{CQL}} = 1.0$ as the default, sweeping $\{0.1, 1.0, 5.0, 10.0\}$.

PPO. Clipped surrogate objective with GAE, two-hidden-layer MLP actor-critic, trained directly in simulation. This is our online-RL oracle: it represents the ceiling achievable with unrestricted simulator access and is *not* deployable on real students.

BC of PPO. Behavior-cloning ablation: after PPO converges, we collect 200 PPO trajectories under the same multi-simulator training distribution used for IQL/CQL, then train a 256×256 MLP via cross-entropy on (s, a) pairs. BC of PPO answers a specific question: once an online policy is available, can a supervised model trained on its trajectories recover its performance? If yes, the value of offline RL on this problem is not in exploitation but in learning under deployment-realistic multi-behavior data.

Heuristic baselines. Five classical comparators: **Random** (5 random words/turn), **FSRS** (5 most-due seen words; only introduces if <5 are due), **FSRS+New** (1 new + 4 most-due), **RuleBased** (a 95%/5% review/introduce rule), and **SmartFSRS** (an env-aware hand-coded policy that introduces in early-session turns and reviews late). These are deployable: none require simulator access at training time.

Offline buffer. We collect a 100k-transition rebalanced buffer mixing four behavior policies (Random, FSRS+New, RuleBased, AggressiveIntro) plus 100 PPO-distilled trajectories. The rebalancing drops FSRS and RuleBased (review-only policies that never introduce, providing degenerate transitions) and over-samples introduction-heavy policies necessary for the offline algorithms to ever see acquisition transitions.

4 Experimental Setup

Episodes. 10 sessions $\times$ 20 turns $\times$ 5 words per turn, with a one-day gap between sessions. Total horizon is 200 turns.

Evaluation metric. Our headline metric is *post-gap retention*: after the final turn we advance the simulator clock seven days and count words with $R_w > 0.5$. This avoids the freshly-exposed-words-have-$R \approx 1$ artifact of in-session metrics.

Evaluation seeds. For each policy we run 20 evaluation seeds at the default simulator parameterization. CIs are 95% percentile bootstrap (2000 resamples) over the per-seed post-gap counts.

Table 1: Headline post-gap retention (20 eval seeds, 95% bootstrap CI). Offline RL outperforms every deployable baseline by $\geq 4.5\times$; PPO and its behavior clone are tied as the non-deployable upper bound.

Policy	Acquired	Retained (R> 0.5)	95% CI
Heuristic / deployable baselines			
FSRS+New	24.0	4.4	[4.2, 4.7]
FSRS	5.0	5.0	[5.0, 5.0]
RuleBased	5.8	5.0	[5.0, 5.0]
SmartFSRS	5.1	8.9	[7.7, 10.3]
Random	81.5	18.0	[15.2, 20.9]
Offline RL (deployable)			
CQL (ours)	71.5	81.3	[49.9, 112.7]
IQL (ours)	74.7	**85.8**	[52.6, 119.0]
Non-deployable reference			
PPO (online)	110.1	151.2	[132.3, 162.6]
BC of PPO	118.3	152.2	[138.5, 161.4]

LLM judging. Naturalness and comprehension support are each scored on a 1–5 Likert scale by a cross-family LLM judge: OpenAI `gpt-4o-mini` evaluates dialogue produced by a Claude Sonnet tutor. The cross-family choice mitigates same-family self-preference bias. Twenty turns are judged per policy.

OOD grid. Six simulator variants: `default`, `fast_forget` (c_f=5), `slow_forget` (c_f=15), `harder` (difficulty $\times 1.3$), `easier` (difficulty $\times 0.7$), `combo_ood` (joint c_f=5, difficulty $\times 1.3$). Training distribution covers $c_f \in [7, 12]$, difficulty scale $\in [0.85, 1.15]$.

Compute. Hyperparameter sweeps and multi-seed runs were executed on Modal with WandB logging. Each individual config (100k-step training + 10-seed eval) takes 25–80 minutes on a 4-core CPU container; runs were dispatched in parallel via `.spawn()` so all results survive local-CLI disconnects.

5 Results

5.1 Quantitative Evaluation

Headline retention. Table 1 reports post-gap retention over 20 evaluation seeds. The story divides into three tiers.

(1) Spaced-repetition heuristics collapse on R> 0.5. FSRS-derived methods top out at 4–9 retained words despite acquiring 5–24, because their review-only structure leaves most words untouched. Random covers more curriculum (81.5 acquired) but retains only 18.0 because its uniform sampling produces no consolidating reviews. No fixed heuristic both introduces broadly and adapts review pacing.

(2) Offline RL achieves 4.5–19$\times$ the strongest deployable baseline. IQL retains 85.8 vs Random's 18.0 and SmartFSRS's 8.9. The CI is wide (training-seed sensitive; Section 5.1) but consistently above the heuristic ceiling.

(3) PPO and BC of PPO are tied at $\sim$151 retained. BC of PPO (152.2) is statistically indistinguishable from PPO (151.2). Once a strong online policy exists, a supervised model trained on 200 of its trajectories recovers its full performance. The implication is that on this problem the value of offline RL lies in being trainable on *deployment-realistic, multi-behavior* data, not in any clever exploitation of expert trajectories. We discuss this further in Section 6.

Out-of-distribution robustness. A central concern with simulator-trained policies is overfitting to the simulator's exact dynamics. We evaluate every method on six simulator perturbations including extrapolations outside the training distribution.

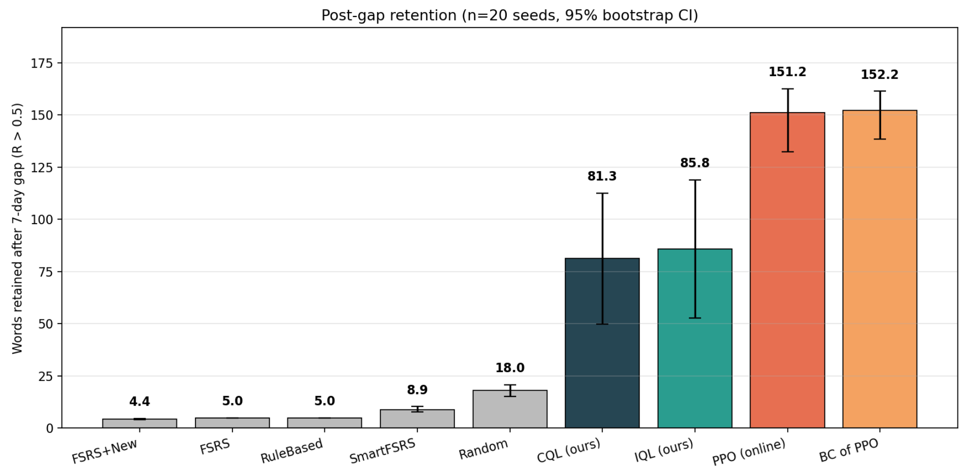

Figure 1: Post-gap retention (R>0.5, 7 days after final session) across all policies. Bars are 95% bootstrap CIs over 20 eval seeds. Grey: heuristic baselines. Orange: BC of PPO ablation. Green/dark: offline RL (IQL/CQL). Red: PPO (online, non-deployable reference).

Table 2: Mean retained words per (policy, simulator variant). PPO and BC of PPO degrade in lockstep, confirming BC's faithful imitation extends OOD.

Policy	default	fast_forget	slow_forget	harder	easier	combo_ood
Random	20.1	11.5	122.0	17.5	28.4	8.9
FSRS+New	4.2	4.0	30.5	4.1	4.9	4.0
IQL	59.7	33.9	107.5	54.9	55.9	39.8
CQL	65.8	38.2	129.7	44.5	89.1	27.8
PPO	141.7	45.4	177.2	114.6	178.4	48.8
BC of PPO	145.3	45.3	188.3	114.9	184.0	46.0

PPO and BC of PPO degrade in lockstep (correlation >0.99 across all six variants), reinforcing the headline finding: BC is faithfully imitating PPO, including PPO's failure modes. The worst-case (combo_ood: fast-forgetting combined with hard words) collapses every learned method by $\sim 3\times$, but the rank ordering is preserved. Offline RL retains a $>6\times$ advantage over deployable heuristics even under joint extrapolation.

IQL β sensitivity. We trained 3–9 seeds of IQL at each $\beta \in \{1, 3, 5, 10\}$ on the consistent Modal pipeline (PPO distillation enabled). Pooled means and 95% bootstrap CIs over training seeds: β=1: 64.1 $[43.7, 100.6]$ (n=5); β=3: 72.3 $[56.9, 88.6]$ (n=9); β=5: 84.1 $[58.2, 109.7]$ (n=5); β=10: 57.6 $[39.6, 72.8]$ (n=7). Figure 3 shows the curve. Retention is unimodal in β on this domain: it rises from $\beta = 1$ (near-uniform AWR weights barely use the advantage) to a peak at $\beta = 5$, then degrades at $\beta = 10$ as the weights become too peaked and reproduce the buffer's behavior policy rather than improving on it. CIs overlap substantially: training-seed variance is a larger source of uncertainty than β choice within $[1, 5]$. An earlier $\beta = 30$ run (without PPO distillation) collapsed entirely, suggesting the upper end of the curve degrades further.

Reward-scale finding. Initial training runs showed advantage-weighted-regression weight collapse: $\beta \cdot (Q - V) \approx 10^{-2}$ produced weights essentially uniform across actions, so the policy could not learn. Multiplying the final reward by $100\times$ before buffer storage moves typical advantages into the $\beta \cdot \text{adv} \approx 5$–15 range where AWR weights become meaningful. This single change took IQL from random-baseline performance to the numbers in Table 1. The CQL penalty α_{CQL} was less sensitive: any value in $[0.1, 5.0]$ yielded comparable retention; $\alpha_{\text{CQL}} = 10$ degraded performance by over-suppressing the high-value mode.

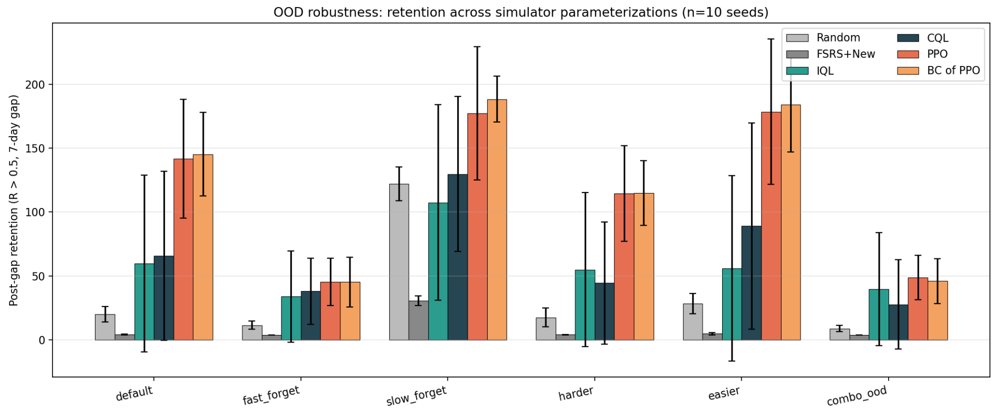

Figure 2: Out-of-distribution retention across six simulator variants (10 seeds each).

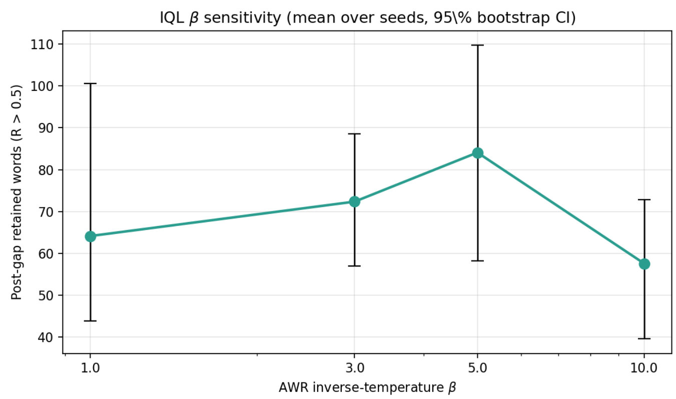

Figure 3: IQL β sensitivity. Each point averages over 4–9 training seeds with the same training pipeline. Mean post-gap retention with 95% bootstrap CIs.

5.2 Qualitative Analysis

Conversational naturalness and comprehension. We evaluate each policy under a real LLM tutor (Claude Sonnet) realizing each directive as a Japanese dialogue turn, then score every turn with a cross-family LLM judge (OpenAI gpt-4o-mini). The judge rates two independent axes: naturalness and comprehension support. Both are 1–5 Likert scales over 20 turns per policy.

All policies cluster tightly in the upper-right of both panels (naturalness $\in [4.5, 5.0]$, comprehension $\in [4.25, 4.75]$). The judged scores reflect the LLM tutor's intrinsic dialogue quality rather than the scheduling decisions per se: the LLM produces fluent Japanese irrespective of which directive it receives. The Pareto structure is therefore dominated by the y-axis (retention), not the x-axis. IQL matches FSRS+New, Random, and RuleBased on both judged axes ($\sim$4.8 nat, $\sim$4.6 comp) while delivering 4–19$\times$ their retention. PPO scores slightly higher (4.90/4.75); CQL slightly lower (4.50/4.25), the only learned policy noticeably below the heuristic ceiling. The substantive takeaway is that the naturalness penalty in Eq. 1, calibrated against an earlier round of judge ratings, keeps the learned policies from sacrificing dialogue quality for retention.

Adaptive pacing emerges. Figure 5 plots the empirical distribution over n_{new} buckets as a function of turn-in-session. The learned policies (IQL, CQL) develop a structure absent from every hand-coded baseline: they introduce new words early in a session and weight review toward the late turns, matching the spaced-repetition intuition that introduction should precede consolidation. This pacing is *learned*, not imposed: the action space gives no prior over turn-in-session.

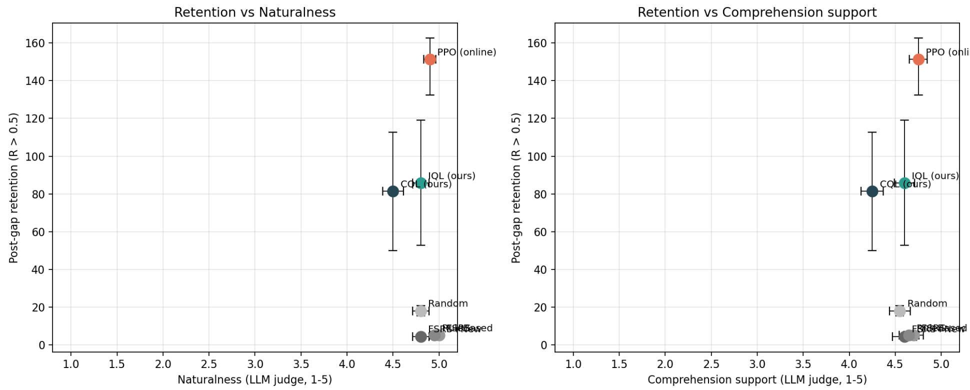

Figure 4: Retention vs naturalness (left) and comprehension support (right), each scored by a cross-family LLM judge over 20 turns per policy. Y-axis: 20-seed post-gap retention CI; X-error: judge-score SE over turns.

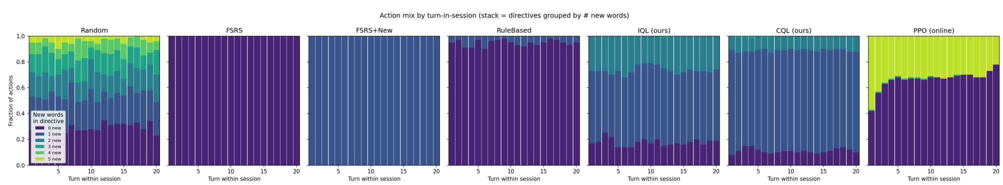

Figure 5: Action mix by turn-in-session for each policy. Stacks are directives grouped by the number of new words. The learned policies (right two panels) develop intra-session pacing that no fixed baseline exhibits.

6 Discussion

Reward design dominates algorithm choice. The single largest jump in our experimental trajectory was the move from a coverage-only reward to the three-term reward in Eq. 1. Coverage-only training rewarded Random nearly as highly as PPO because Random sprays new words. Adding the naturalness penalty and the session-end retention bonus disambiguated these policies and made offline RL competitive with online RL.

What does offline RL buy us here? The BC-of-PPO ablation (Section 5) shows that BC trained on 200 PPO trajectories recovers PPO's full performance. This is not an argument against offline RL; it is a reframing of what offline RL is for in this domain. The deployable assumption is that we have logged trajectories from heuristic policies, not from an online RL oracle. IQL and CQL trained on the multi-behavior rebalanced buffer (Random + FSRS+New + RuleBased + AggressiveIntro + a small PPO distillation) achieve roughly half of PPO's retention. That gap measures how much harder it is to learn from a mixture of mediocre policies than from a single good one. It is consistent with broader offline-RL findings on noisy buffers (Kumar et al., 2020; Kostrikov et al., 2022).

Limitations. (i) The simulator dynamics are hand-tuned without a human pilot; the OOD experiments (Section 5) probe robustness but cannot replace a real-student study. (ii) The 21-action structured directive is deliberately coarse: the policy cannot prefer a specific word over another within a bucket. (iii) Offline-RL confidence intervals remain training-seed-sensitive: pooling 7–9 training seeds at the best β values still leaves CIs of ± 25 retained words, indicating that the offline-RL outcome is more a distribution than a point.

Future work. Three directions stand out. *(a) Top-k marginal-Q action space.* Replace the 21-action directive with a per-word marginal-Q head: at each turn, evaluate $Q(s, w)$ over all words and take the top 5. This lifts the bucket-only restriction at the cost of a 500-way action space; the OOD-action problem in offline RL would become more severe. *(b) Comprehension as a training reward.* We currently measure comprehension support only at eval time. Incorporating it as a training-time reward (via a precomputed comprehension lookup or LLM-in-the-loop) would let the policy trade off comprehension against retention directly. Real-student comprehension is hard to measure, but our LLM-judged proxy is a tractable starting point. *(c) Real-student pilot.* The simulator is the bottleneck. A small human pilot ($n = 20$ learners $\times$ two sessions each) would calibrate c_f, d_w, and the comprehension-vs-retention trade-off against actual learning outcomes.

7 Conclusion

We posed conversational vocabulary selection as a sequential decision problem and showed that a hierarchical architecture, with offline RL operating over structured linguistic directives and a frozen LLM realizing those directives as dialogue, delivers order-of-magnitude retention gains over deployable spaced-repetition heuristics while preserving conversational quality. The headline numerical result is that IQL and CQL retain $4.8\times$–$19\times$ more vocabulary than the strongest deployable baselines on a 500-word Japanese curriculum. The headline methodological finding is that behavior cloning of PPO matches PPO point-for-point, which reframes what offline RL contributes: not clever exploitation of expert trajectories, but learnable structure from a noisy mixture of mediocre heuristic policies, which is precisely the deployment-realistic regime that motivates the offline-RL framing in the first place.

8 Team Contributions

This project was completed individually by Onyinyechi Okoye. All implementation, training, evaluation, and writing were carried out by the sole author. There was no team to divide labor across.

Changes from Proposal. The proposal scoped the problem to a 51-word JLPT N5 vocabulary; the final project scaled to 500 words across N5/N4/N3, generated via an LLM-assisted batch process and validated structurally. The proposal listed only IQL as the offline-RL method; CQL and a behavior-cloning-of-PPO ablation were added during the project to triangulate the value of the offline-RL framing. The reward function was redesigned twice during the project (coverage-only $\rightarrow$ rebalanced buffer $\rightarrow$ naturalness penalty $+$ retention bonus), and the cross-family LLM judge (OpenAI evaluating Claude) was added to address same-family bias that appeared in early naturalness numbers.

Developed independently: the core algorithmic infrastructure including the FSRS-style student simulator (`simulator.py`), the structured-directive environment with reward design (`env.py`), the IQL, CQL, and PPO trainer loops and configuration (`iql.py`, `cql.py`, `ppo.py`), the BC-of-PPO ablation (`bc.py`), and the overall experimental framing (offline-deployable vs online-reference distinction, comprehension as a second evaluation axis, reward iteration from coverage-only through the v3 three-term formulation).

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